Longitudinal Modeling of Sleep Timing, Duration, and Social Jetlag Across the Adult Lifespan Using High-Resolution Wearable Data

All of Us Research Program Fitbit sleep phenotyping

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Presenter: Diego R. Mazzotti, Ph.D. (on behalf of Mason Manetta)

42,290	22.4M	530

Disclosure Information

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sleepmeeting.org

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Wearables and Longitudinal Monitoring of Sleep and Circadian Traits

Why this matters

- Sleep is important to health and chronic disease risk.
- Most large epidemiologic studies rely on self-report or short-term monitoring.
- Wearables create scalable, repeated measurements of sleep timing, duration, and variability.

Longitudinal Fitbit data move sleep phenotyping from sparse recall to repeated behavioral measurement.

Published context

Prior All of Us Fitbit analyses have linked wearable-derived sleep measures to clinical phenotypes, motivating higher-resolution longitudinal modeling of sleep behavior.

Core outputs

Onset	sleep start time
Offset	wake time
Midpoint	center of sleep episode
Duration	sleep window / time asleep

Sources of Variability

Extrinsic

- Weekdays vs. weekends / free days
- Seasonality and holidays
- Environmental exposures: photoperiod, temperature, humidity, daylight saving time

Intrinsic and social

- Age and sex
- Employment status and social scheduling constraints
- Stable participant-level differences captured by repeated measures

These factors are established individually, but longitudinal wearable data allow them to be modeled together within the same participants.

Study Aim

Characterize demographic and environmental contributors to longitudinal variability in sleep patterns

using device-measured Fitbit sleep data
from the All of Us Research Program.

Onset
sleep start

Midpoint
sleep timing

Duration
sleep amount

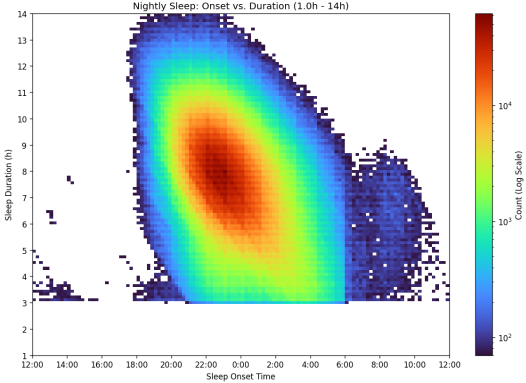
All of Us Research Program and Fitbit Sleep

All of Us

- National longitudinal cohort with EHR, survey, genomic, and wearable data.
- Participant-level covariates include demographics, employment, BMI, ZIP3 geography, and socioeconomic context.

Fitbit measurements

- Sleep daily summaries provide nightly sleep duration.
- Sleep-level records provide high-resolution onset/offset



Filtered person-night onset and duration density.

Participant Characteristics

Characteristic	Valu
Included participants	42,29
Filtered sleep nights	22,428,61
Age	52 ± 17 y
Female	67.4%
Male	32.3%
Nights recorded	530 ± 73
Nightly duration	447 ± 65 mi
Nightly duration	7.5 ± 1.1 h

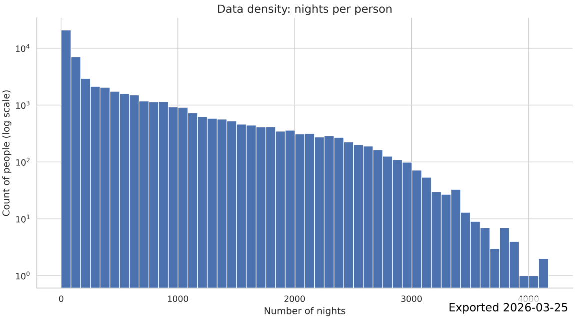
AoU participants with Fitbit sleep

↓

valid primary sleep episodes

↓

42,290 participants with
22.4M filtered nights

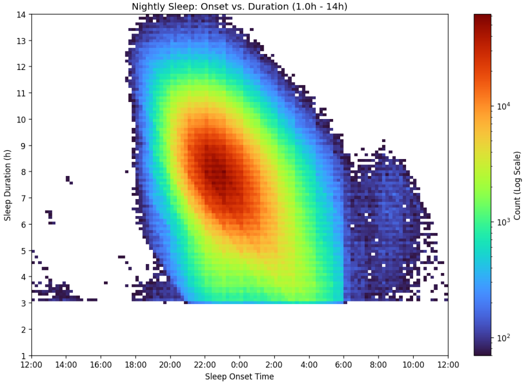


Distribution of nights per participant.

From Fitbit Records to Nightly Sleep Episodes

Filtering and episode construction

- Sleep-level records linked to daily summaries.
- Non-primary sleep excluded.
- Implausible raw durations excluded: > 18h, zero/negative, common sensor artifact values.
- Segments clustered into nightly episodes; largest main sleep cluster retained per logical night.
- Sleep date assigned using a 6-hour backward shift.



Onset-duration density after filtering shows expected overnight sleep structure.

Modeling Strategy

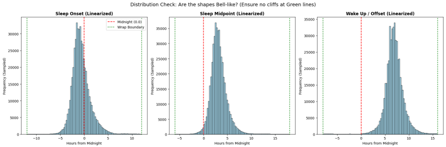
Linear mixed-effects models

$$Y_{ij} \sim \mathrm{poly}(\mathrm{age}_{ij}, 2) + \mathrm{weekend}_{ij} \times \mathrm{employment}_i + \mathrm{sex}_i +$$

- Outcomes: onset, offset, midpoint, duration.
- Person random intercept controls stable between-person differences.
- Quadratic age captures non-linear lifespan trends.
- Employment-by-weekend interaction estimates social constraint/social jetlag patterns.

Time handling

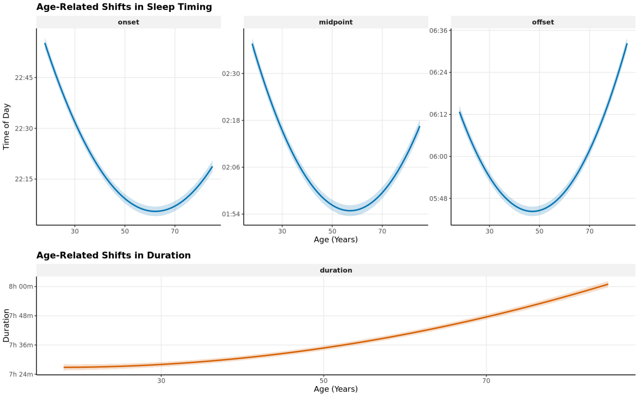
- Timing variables were linearized because distributions were primarily unimodal.
- Clock times before noon shifted forward by 24h.
- Example: 1:00 AM $\rightarrow$ 25.0; 11:00 PM $\rightarrow$ 23.0.



Sleep and Circadian Traits x Age and Sex

- Age showed a quadratic association with sleep timing across the adult lifespan.
- Model-estimated sleep midpoint advanced by approximately 42 minutes from age 18 to age 57.
- Midpoint: 2:37 AM at age 18 vs. 1:54 AM at age 57.

A simple linear age term would miss the lifespan shape of sleep timing.

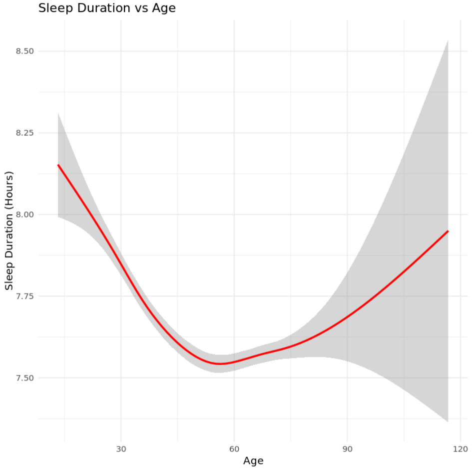


Sleep Duration Across the Adult Lifespan

- Duration increased with age in adjusted models.
- Estimated gain: >34 minutes between ages 18 and 85.
- Age 18: 7.45h; age 85: 8.02h.

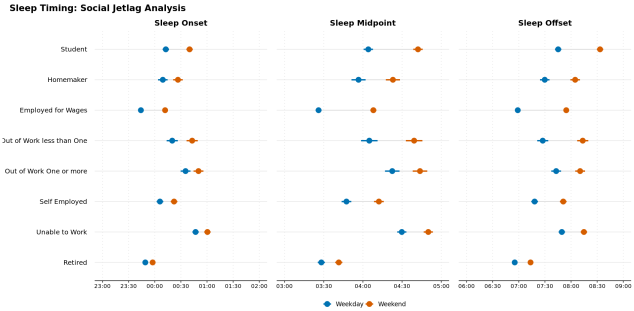
Why it matters

Timing and duration change together, so future analyses should consider mutually adjusted timing-duration phenotypes.



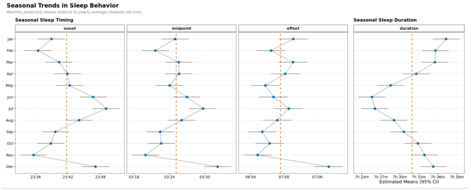
Sleep and Circadian Traits x Other Sociodemographics

- Weekend status delayed midpoint, onset, and offset across employment groups.
- Participants employed for wages showed the largest midpoint delay.
- Employed for wages: weekday midpoint 3:26 AM vs. weekend midpoint 4:08 AM.
- Weekend sleep duration extended by approximately 28 minutes.



> Social timing constraints are visible in wearable sleep at pop-

Sleep and Circadian Traits x Environmental Variables



- Month terms indicated seasonal differences in sleep behavior.
- Sleep duration was approximately 11 minutes lower in June than the January maximum.
- Unadjusted weekly trends suggest non-sinusoidal structure, likely mixing photoperiod, calendar behavior, holidays, and DST.

What This Adds

Population-level characterization

- Quantifies non-linear age effects in device-measured sleep.
- Separates workday/free-day structure from participant-level baselines.
- Provides adjusted estimates across employment categories.

Foundation for next-stage analyses

- Genetic studies of sleep timing and chronotype.
- Clinical risk stratification using longitudinal sleep phenotypes.
- Environmental extensions: photoperiod, temperature, humidity, DST.

Large-scale wearable data can turn sleep timing from a sparse self-report phenotype into a longitudinal, model-ready exposure.

Limitations and Interpretation Guardrails

Measurement

- Fitbit-derived sleep depends on device classification of main sleep.
- Shift work is not directly measured; employment status is a proxy for social schedule constraints.
- Linearization works for the observed distribution but may under-represent true night-shift phenotypes.

Cohort and covariates

- AoU Fitbit participants are not a probability sample of U.S. adults.
- Pregnancy episode exclusion was not feasible in the V8 data context.
- ZIP3 geography is coarse for environmental and DST classification.

Conclusions and Future Directions

1. Sleep timing and duration are dynamically structured by non-linear age effects.
2. Employment and weekend status reveal strong social timing constraints, including measurable social jetlag.
3. Seasonal patterns persist at population scale and motivate environmental modeling.

Longitudinal wearable sleep data provide a scalable foundation for personalized circadian epidemiology.

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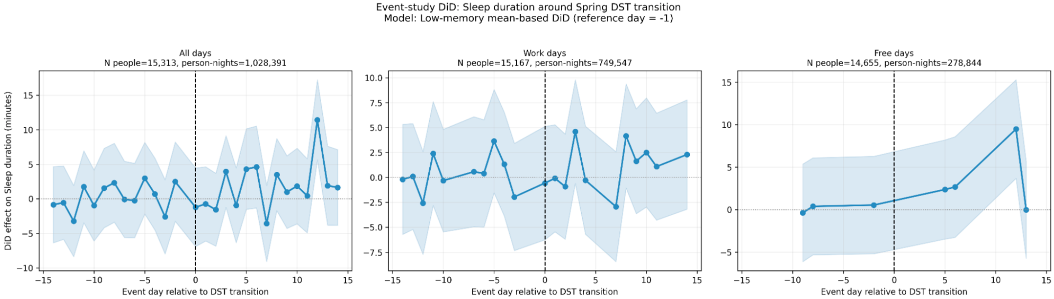
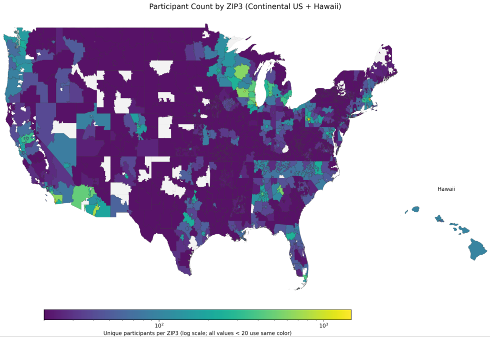
Questions and discussion

Backup: Validation Against Published AoU Sleep Structure



Reference comparison with published All of Us Fitbit sleep structure; used during extraction validation.

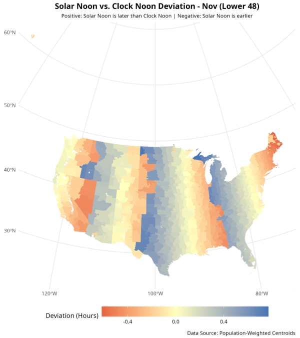
Backup: Geographic and Environmental Extensions



Participant counts by ZIP3.

- ZIP3 enables coarse geography for SES, DST, photoperiod, and PRISM weather linkage.
- Future models can estimate temperature, vapor pressure deficit, precipitation, and photoperiod effects.
- Environmental analyses should account for ZIP3 population weighting and non-random participant geography.

Backup: Daylight Saving Time Design



DST contrast

- Classify ZIP3 as DST-observing vs. non-DST control.
- Center each person-year using pre-transition baseline days -14 to -7.
- Estimate event-day differences around spring/fall transitions.